COUNTRY PROFILES

# Nuclear Power in Czech Republic

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- The Czech Republic has six nuclear reactors generating about one-third of its electricity.
- KHNP is to construct two large reactors at Dukovany with construction starting before the end of the decade.
- Its first commercial nuclear power reactor began operating in 1985.
- Government commitment to the future of nuclear energy is strong.

Privacy - Terms

Policy calls for the share of electricity provided by nuclear to rise to 68% by 2040.

6

OPERABLE REACTORS

4,212 MWe

0

REACTORS UNDER CONSTRUCTION

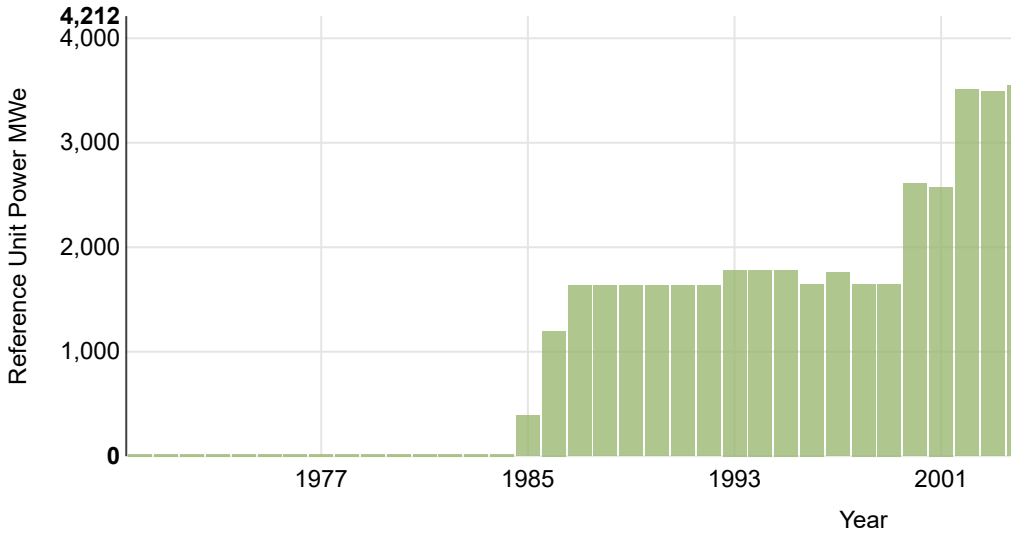
0 MWe

0

REACTORS SHUTDOWN

0 MWe

## Operable nuclear power capacity




# Electricity sector

**Total generation (in 2024):** 73.9 TWh

**Generation mix:** nuclear 29.7 TWh (40%); coal 26.8 TWh (36%); biofuels & waste 5.5 TWh (7%); natural gas 3.8 TWh (5%); hydro 3.6 TWh (5%); solar 3.6 TWh (5%); wind 0.7 TWh (1%).

**Import/export balance:** 6.7 TWh net export (15.3 TWh imports; 22.0 TWh exports)

**Total consumption:** c. 55 TWh

**Per capita consumption:** c. 5200 kWh

*Source: International Energy Agency and The World Bank. Data for year 2024.*

# Nuclear power industry

### Operable reactors in Czech Republic

| Reactor Name | Model | Reactor Type | Reference Unit Power (MWe) | Construction Start | First Grid Connection |
|---|---|---|---|---|---|
| Dukovany 1 | VVER V-213 | PWR | 510 | 1979-01 | 1985-02 |
| Dukovany 2 | VVER V-213 | PWR | 510 | 1979-01 | 1986-01 |
| Dukovany 3 | VVER V-213 | PWR | 510 | 1979-03 | 1986-11 |
| Dukovany 4 | VVER V-213 | PWR | 510 | 1979-03 | 1987-06 |
| Temelin 1 | VVER V-320 | PWR | 1086 | 1987-02 | 2000-12 |
| Temelin 2 | VVER V-320 | PWR | 1086 | 1987-02 | 2002-12 |

## Nuclear industry development

In 1958, the Czechoslovak government started building its first nuclear power plant – a gas-cooled heavy water reactor at Bohunice (now in Slovakia). This 110 MWe net Bohunice A1 reactor, built by Skoda, was completed in 1972 and ran until 1977. Construction on four VVER-440 reactors at Bohunice commenced

in the 1970s and the units were put into operation by the mid-1980s (see page on Nuclear Power in Slovakia).

In 1978, construction commenced on the Dukovany plant – the first nuclear plant in what is now the Czech Republic. The four VVER-440 model V-213 reactors were designed by Russian organizations and Energoprojekt and built by Skoda Praha. These entered commercial operation in 1985-87 and have been upgraded since.

In 1982, work started on the Temelin plant (also in what is now the Czech Republic), designed by Russian organizations and Energoprojekt and built by VSB[c] with engineering by Skoda Praha. Planned as a four unit VVER-1000 model V-320 plant, full construction on units 1&2 commenced in 1987. However, following the Velvet Revolution of 1989, the new democratic government decided in 1990 to suspend construction on units 3&4. At the time there were several protests against the plant and some political wavering over the plant's future, although construction continued on units 1&2 to some extent. Then, with the splitting of Czechoslovakia, the new government of the Czech Republic formally decided in March 1993 to complete units 1&2. Following a call for tenders, Westinghouse was chosen to replace the entire instrumentation and control (I&C) systems – the first time that Western digital I&C was integrated with Russian reactor technology[d] – and the radiation monitoring and diagnostic systems[1]. In addition, Westinghouse was to supply the fuel (initial core and four reloads). The reactors started up in 2000 and 2003, with the upgrading having been financed by operator CEZ with a loan from the World Bank. Commercial operation was in June 2002 and April 2003, respectively.

Heat from Temelin is supplied to local towns Tyn and Vltavou 5 km away, and in 2020 CEZ began building a connection to the city of České Budějovice 24 km away, which will cover 30% of the city's needs. An environmental impact assessment for a project to pipe heat 40 km from Dukovany to Brno was lodged in July 2010, but did not proceed. However, the project was revived following Russia's offensive in Ukraine in 2022 in order to reduce the country's reliance on Russian-supplied gas.

CEZ a.s., which owns and operates both the Dukovany and Temelin plants, is almost 70% state-owned[e].

## Nuclear energy policy

A 2011 draft national energy policy to 2060 involved a major increase in nuclear power, to 13.9 GWe or up to 18.9 GWe in the case of major adoption of electric vehicles. It would then provide some 60% of the country's power. The version adopted in November 2012 assumed at least 50% from nuclear, with two new reactors being built at Temelin and one at Dukovany to take production to 46.5 TWh by 2025, and 55.2 TWh/yr later, hence further nuclear sites should be identified and prepared. The four reactors at Dukovany would get 20-year operating lifetime extensions, to 2045-47. Coal use would decrease to one-third of 2012 levels by 2040, gas-fired power would increase, and support for renewables would reduce dramatically by 2014. Nuclear plants were to furnish district heating to Brno and other cities by 2030.

The 2015 national energy policy reiterated most of the plans outlined in 2012, with one new reactor at Dukovany and two at Temelin, but without any state guarantee on electricity prices. The policy is explicitly part of the country's commitment to a European Union target for cutting carbon emissions. In the plan new nuclear capacity of 2500 MWe is to be added by 2035, and more thereafter, making nuclear the main source of electricity production with its share rising from 35% to between 46% and 58% in 2040. The plan calls for four years' fuel reserve.

In a speech in December 2019 the Czech prime minister stated that construction on the country's next unit – at Dukovany – was expected to begin in 2029, with the reactor in operation by 2036, bringing nuclear's share of electricity to 40% by 2040.

Earlier in January 2016 the government set up a new committee headed by the prime minister to coordinate the development of nuclear power in the country. A new nuclear envoy is to serve as the main coordinator for these developments. The committee will be responsible for new construction, supply chain, wastes, and legislation to move the nuclear sector forward. In January 2017 three working groups were set up to give new impetus to the nuclear component of the 2015 energy policy, initially a new reactor at Dukovany.

In early 2024 the Czech Republic invited vendors to bid for the construction of four reactors (see below). In January 2025 the government submitted its updated National Energy and Climate Plan to the European Commission, which states that the share of generation provided by nuclear would rise to 68% by 2040.

## Plant uprates and lifetime extension

### Dukovany

All four Dukovany units were uprated from 440 to 456 MWe gross over 2005-08 by replacing low-pressure turbines. This was the first part of a programme to boost Dukovany's gross output by about 240 MWe by 2012. The 38 MWe uprates of units 3&4 brought their gross capacities to 525 MWe – from improved fuel, replacing the high-pressure turbine, refurbishing the generator, and I&C changes. Similar uprates of units 1&2 followed, all completed by the end of 2012.

At the beginning of 2009, CEZ commenced its long-term operation (LTO) project, the immediate focus of which was to extend the planned operating lifetimes of the Dukovany reactors by 10 years. The project consisted of some 230 sub-projects and cost CZK 18 billion ($825 million) from 2009 to 2017. Austria, whose border is 50 km away, objected to the proposed extensions, but in March 2016 unit 1 was licensed for continued operation indefinitely subject to ongoing reporting, followed by unit 2 in July 2017 and units 3&4 in December 2017.

In June 2020 CEZ stated that it expects to invest about $2.3 billion over the next 27 years to extend the operating lifetimes of the four reactors at Dukovany by a further 20 years to a total of 60. The work would focus on reactor management and control systems, cabling, and safety systems.

In February 2023 CEZ announced that it planned to invest around $100 million into Dukovany 4 to extend the plant's operating lifetime to 60 years, to 2047. The same month, a contract was signed with Framatome to install new Spinline testers at the Dukovany units – a modular digital solution for I&C safety systems designed to implement the plant's safety functions.

In March 2025 CEZ said that the four units at Dukovany were capable of supplying 2048 MWe, almost 300 MWe more than at grid connection. In July 2025 CEZ said that it was scoping out the modernization of turbine halls at Dukovany with a view to increasing output and operation to 60 years.

### Temelin

With Temelin 1&2, Skoda Power modified the high pressure turbines over 2004-07 in a €26 million project to achieve 1013 MWe gross, 963 MWe net. Further uprates have taken both units to 1082 MWe gross. In September 2020 the State Office for Nuclear Safety (SUJB) confirmed a 10-year operating licence renewal for unit 1. Unit 2 is undergoing the same renewal process.In July 2021 CEZ announced that replacement of the two moisture separator reheaters had increased the capacity of Temelin 1 to 1125 MWe.

In September 2025 Doosan Škoda Power was awarded a contract to supply new generators to the Temelín plant.

## New nuclear capacity: Temelin 3&4, Dukovany 5&6

The 2004 state energy policy envisaged building two or more large reactors, probably at Temelin, eventually to replace Dukovany. In July 2008, CEZ announced a plan to build two more reactors at Temelin totalling up to 3400 MWe, with construction start in 2013 and commissioning of the first unit in 2020. In mid-2008, CEZ asked the Environment Ministry for an environmental assessment for the new units. A public tender process for contractors to build the two new reactors at Temelin commenced August 2009. At the time, CEZ quoted a March public opinion poll showing 77% of citizens (and 56% of Green party voters) supporting the new Temelin units[3].

In March 2010, CEZ announced that discussions had begun with three candidates prior to the bid submission. The three vendor groups were: a consortium led by Westinghouse with 1230 MWe gross (1140 net) AP1000; a Škoda JS/Atomstroyexport/OKB Gidropress consortium[f] with AES-2006/MIR-1200 of 1158 MWe gross (1078 net); and Areva with 1750 MWe gross (1650 net) EPR. Bids were formally invited by CEZ in October 2011 for supply of two complete nuclear power plant units on a "full turnkey basis, including nuclear fuel supply for nine years of operation." The designs proposed were to be licensed in the vendors' home countries or in a EU member state, and were to comply with Czech and EU requirements and safety requirements defined by the IAEA and the Western European Nuclear Regulators Association.

Bids for Temelin were submitted in July 2012, and the contract was to be signed late in 2013 but was then deferred for about 18 months, to mid-2015 and following completion of a new energy strategy by the new government. Rosatom, through Rusatom Overseas, offered full vendor financing, though said it would prefer 49%. Areva and Westinghouse originally said that they were not interested in any financing or operational aspects, but in mid-2013 the US Export-Import Bank offered to lend CEZ half the cost of the plant if it used Westinghouse technology. The loan would have been for 25 years at one percentage point above US 10-year treasury bonds. CEZ said it would seek a strategic partner with which to share the risk of the project, following the choice of reactor technology.*

* In October 2012 CEZ said that the Areva bid was disqualified because it did not meet all the statutory requirements under the Czech Republic's Public Procurement Act and that the company had “not fulfilled some other crucial criteria defined in the tender.'' Areva contested the decision but its appeal was rejected by the Anti Monopoly Office.

In November 2012 CEZ applied to SUJB to build two new units at Temelin. In January 2013 the government gave environmental approval for the two units. (A 5000-page environmental impact statement had been lodged with the government in May 2010.)

The previous government was planning to legislate for a cost-difference guarantee for electricity from Temelin 3&4 to ensure that investment was viable. This would cover the difference between wholesale electricity prices and price levels needed to cover construction costs. The Ministry of Industry and Trade wanted it written into a new long- term Czech energy framework, but this was opposed by the Ministry of Finance. Estimates of its impact vary up to 10% additional on retail power bills. The Industry Ministry was working on €60/MWh, others suggest that €90 would be needed, indexed. CEZ required €70/ MWh for the new units to be profitable, compared with mid-2013 forward prices of under €40. The prime minister of the coalition government elected in 2013 said it was not open to providing price guarantees that would “dramatically burden” consumers, after its experience in support of renewable sources, notably solar PV. CEZ responded by saying that it needed some long-term price assurance or it would abort the project and leave consumers uncertain about supply and prices in the 2020s.

In March 2014 the Ministry of Trade & Industry said that it was considering forming a wholly state-owned company to build

Temelin 3&4 and then lease them to CEZ, this being seen as more practical than the UK policy of long-term pricing involving cost-difference guarantee, but it would transfer financial risk to the state budget.

In April 2014, following government confirmation that it would not provide any future price guarantees, CEZ informed Westinghouse, the Russian-Skoda consortium and Areva that it had cancelled the procurement process in accordance with the Public Procurement Act. CEZ was reported to be favoured as investor.

New bids were expected in 2015. Korea Electric Power Co (KEPCO) announced its intention to bid, and in December 2015 a nuclear cooperation agreement was signed with South Korea. Earlier in August 2014 China's deputy prime minister expressed interest in the project on behalf of his country. In September 2014 Westinghouse offered to the Czech ministries of finance and industry & trade the possibility of vendor equity as in the UK, where parent company Toshiba had 60% equity in the Moorside AP1000 project, now aborted.

In June 2015, the cabinet approved a Ministry of Trade and Industry long-term plan for the nuclear industry, involving one new unit at Dukovany, and possibly three more at the two sites. It recommended that CEZ create a subsidiary company to prepare construction plans and explore options for financing the new reactors.

In mid-2016 CEZ asked the Environment Ministry for an environmental assessment for two new units at Dukovany. The government's national action plan for nuclear energy adopted in June 2015 said that Dukovany 5 had priority over Temelin in order to maintain production at the site after the old reactors are retired about 2047. In October 2016 Rusatom Overseas submitted to the government and CEZ an offer to build a VVER-1200 reactor at Dukovany. The company was prepared to offer a wide range of financial and commercial terms including an EPC contract and holding minority ownership in the unit, similar to the (now cancelled) Finnish Fennovoima arrangement. Five other companies also submitted offers: EDF/Areva, Atmea, China General Nuclear Group, Korea HNP and Westinghouse.

Though the government had not yet resolved the financing question, early in 2017 CEZ held talks with six companies and consortia which had expressed interest in building reactors at Temelin and Dukovany: Westinghouse; Rusatom; EDF and Areva; Atmea (Areva-Mitsubishi Heavy Industries joint venture); China

General Nuclear Power Corp (CGN); and Korea Hydro and Nuclear Power (KHNP). In February 2020 Atmea withdrew from the process. In March 2021 the Ministry of Industry & Trade selected EDF, KHNP, Rosatom and Westinghouse to participate in a pre-qualification round for a tender for Dukovany unit 5. However in April 2021 the Czech Republic announced that it would exclude Russia and China from the tender. The Chinese embassy protested at CGN's exclusion and Rosatom described its exclusion as "politically motivated". In January 2022 Westinghouse signed MoUs with seven companies in the Czech Republic, covering the potential deployment of the AP1000 as unit 5 of Dukovany.

Earlier in July 2019 the Czech government published a resolution outlining its intention to provide loan guarantees to CEZ to allow it to secure cheaper financing. However, it confirmed it would not provide a return guarantee for investors in the project. In May 2020 the Czech government agreed to a state loan covering 70% of the costs of building the new reactor. The loan would initially be interest free, followed by a minimum rate of 2% (or 1% above the state loan at the time of repayment) once the plant begins operation.

In March 2020 CEZ announced that it had submitted a licence application for two new PWRs up to 1200 MWe each at Dukovany. In June 2021 it applied for a zoning permit for up to two new units, having prepared 30,000 pages of documentation.

In July 2020 the Czech cabinet approved a proposed new law which would allow the government and CEZ to agree a minimum 30-year power purchase agreement (PPA) for Dukovany 5. The price should allow CEZ to recoup the investment cost and make a profit. The state would sell the electricity into the wholesale market. Electricity consumers would be faced with surcharges on their bills to make up for losses if wholesale power prices are lower than the guaranteed price or benefit from lower electricity bills if the prices are higher, as with the UK contract for difference system. There would be a cap on the possible surcharge. The PPA is framed to suit follow-up projects over 100 MWe coming online after 2030 including small modular reactors.

In March 2022, wholly-owned ČEZ subsidiary Elektrárna Dukovany II launched a tender for the construction of a new nuclear power plant in Dukovany. At the end of November 2022 CEZ announced that it had received initial bids from EDF, Westinghouse and Korea Hydro and Nuclear Power (KHNP).

In October 2023 EDF submitted a bid to build an EPR reactor for CEZ's planned Dukovany 5. Previous reports suggested that EDF

would offer its EPR1200, a 1200 MWe PWR adapted from the EPR design.

The tender was originally for a binding offer for one new unit at Dukovany and non-binding offers for up to three more – another one at Dukovany and two at Temelín. But in February the Czech government announced it was changing the tender to binding offers for up to four new units, citing the cost savings per unit if they were not procured on a unit-by-unit basis.

In January 2024 the government invited EDF and KHNP to provide binding bids that include costs for the construction of up to four reactors. In June 2024 CEZ submitted its evaluation of the bids received from EDF and KHNP to the Czech government. In July 2024 the Czech government announced that KHNP had won the contest, and said that contract negotiations with the Korean company would begin. The Czech government said that the KHNP bid was for a cost of around CZK200 billion per APR-1000 unit ($8.6 billion in July 2024).

In May 2025, the government announced that it would purchase an 80% stake in the Elektrárna Dukovany II project, with CEZ retaining a 20% stake. The 80% stake was valued at $163 million. In October 2025 the government requested European Commission approval to finance Dukovany 6, following approval for Dukovany 5 in April 2024, as the government now plans to build both units simultaneously. A decision is expected by late 2026.

Site survey work began for Dukovany 5&6 in August 2025.

### Planned and proposed reactors

| Reactor | Model | MWe | Construction start | First power |
|---|---|---|---|---|
| Dukovany 5&6 | APR-1000 | 1050 | 2029 | by 2036 |
| Temelin SMR | Rolls-Royce SMR | 470 | | |
| **Total planned (3)** | | 2570 | | |
| Temelin 3&4 | EPR, APR-1000 or AP1000 | 1200 approx each | | |
| Tušimice SMR | Rolls-Royce SMR | 470 | | |
| Prunerov SMR | Rolls-Royce SMR | 470 | | |
| **Total proposed (4)** | | 3340 | | |

CEZ also has a 49% share of a joint venture with Slovak state-owned Javys[g] to build a new reactor at Bohunice in Slovakia, to be known as the New Bohunice Block. However, this project is stalled.

**Small modular reactors**

In February 2020 CEZ signed a memorandum of understanding (MoU) with GE Hitachi to examine the feasibility of constructing a BWRX-300 – a small modular reactor (SMR) of 300 MWe – in the Czech Republic. This followed the signing in September 2019 of an MoU with NuScale to explore SMR deployment in the country. The capacity of the NuScale SMR has since been increased to 77 MWe. In November 2020 CEZ signed a MoU with Rolls-Royce to explore the potential for deploying its 400 MWe (since increased to 470 MWe) SMR in the Czech Republic. CEZ has also signed MoUs relating to the deployment of SMRs with EDF, Westinghouse, Holtec and KHNP.

In January 2020 the nuclear research institute, UJV Rez, patented a fluoride high temperature SMR design named Energy Well, with a 20 MWt capacity and a seven-year fuel cycle. In May 2021 UJV Rez announced its project to design by 2035 a high-temperature gas-cooled reactor (HTR) aimed at the heating and industrial sectors. This HeFasto project is to develop a 200 MWt reactor operating at up to 900 °C. Three versions will be offered for heating, cogeneration and the chemical industry. It proceeds from the Allegro project to develop a 50-100 MWt gas-cooled fast reactor, also supported by Hungary, Poland and Slovakia, focused on the V4G4 Centre of Excellence set up in Slovakia in 2015.

In March 2022 CEZ said that it had earmarked an area of land at its Temelin site to be used in the future for the construction of the country's first SMRs. In February 2023 CEZ identified the coal-fired power plants at Dětmarovice and Tušimice as the preferred locations for a second and third SMR, with CEZ hoping to have the sites in operation by the second half of the 2030s. In November 2023 the government approved the Czech SMR Roadmap, which sets out the framework for the potential for advanced SMRs in the Czech Republic. The roadmap includes information on the various design options and identifies 45 potential host sites.

In April 2025 US-based engineering company Amentum was selected to conduct environmental impact assessments related to the deployment of Rolls-Royce SMRs at Temelin and Tušimice.

In June 2025 the government confirmed plans to build SMRs in Tušimice and Prunerov in northern Bohemia, which both host coal-fired power plants.

In July 2025 an early works agreement was signed between CEZ and Rolls-Royce SMR to start site-specific work at Temelin.

# Fuel cycle

## Mining

Czech uranium mining, which once provided over 2500 tonnes of uranium (tU) per year, declined rapidly from 1990 following the collapse of the Communist government. A total of 112,055 tU was produced 1946-2022. By 1994, Czech production had fallen to 600 tU/yr, and through the last decade production declined steadily to 138 tU in 2016, with no production since.

The last underground mine – the Roun mine on the Rožná deposit in the Dolní Rožínka locality 50 km northwest of Brno in southern Moravia – operated by state-controlled Diamo, was due to close in 2003, but was extended due to rising uranium prices[h].* In 2014 the mine was substantially depleted and after production ceased at the end of 2016, the government closed it early in 2017. Rozna was the last operational uranium mine in central Europe.

Closure of Rozna reduced production to a few tonnes per year from the former ISL operation at Hamr-Stráž pod Ralskem[i] in northern Bohemia. At the time a major (€1.25 billion) groundwater rehabilitation project was under way at Hamr-Stráž pod Ralskem following 32 years of mining. In 2009 Diamo proposed recommencing uranium mining at Stráž pod Ralskem, but the government rejected the idea.

Diamo also completed a feasibility study on reopening the 1980s Brzkov mine near Jihlava in southern Moravia about 50 km northwest of Rozna, which the Industry Ministry said contains about 3150 tU reserves some 300 m deep. Diamo said it would take 6-7 years to properly commission the mine to produce 210 tU/yr, but development would depend on uranium prices.

* In 2007, Australian company Uran Ltd made an offer to participate in the Rozna mine, but this was rejected in favour of local plans. Diamo confirmed it would explore expanding the mine below the current 1200 m depth and, should it be successful in defining further reserves, Uran Limited would be offered potential participation[5]. Uran also applied for four uranium exploration permits at locations near Rozna and two near Stráž pod Ralskem,

but these applications were unsuccessful due to political opposition[6]. The company has since abandoned uranium.

There are about 14 other historic uranium mining sites.

There are still substantial reserves of uranium (around 160,000 t)[10] in sandstone-hosted deposits in areas surrounding Stráž pod Ralskem. However, they are located underneath a major drinking water aquifer. According to the 2017 Raw Materials Policy, the next 25-30 years should be used for research of new mining and processing methods, which would not cause irreversible damage to the environment.

All the uranium mining and remediation activities in the Czech Republic are the responsibility of the Diamo state enterprise.

## General

Most fuel has been sourced within the country, with conversion, enrichment and fuel fabrication being undertaken in Russia by TVEL. Fuel for Dukovany is supplied by TVEL. In July 2019 a contract was signed with TVEL to supply Dukovany with its new VVER-440 fuel, RK 3+. Fuel for Temelin was initially supplied by Westinghouse but, in 2006, TVEL won a contract to supply fuel for the two VVER-1000 reactors for 10 years from 2010.[7] In 2016 Westinghouse was contracted to supply six test assemblies to be evaluated over two years. In 2018 the Czech Office for Nuclear Safety approved the use of the assemblies, and these were loaded in April 2019. In April 2022 CEZ announced that it had selected Westinghouse and Framatome to provide fuel for Temelin. Contracts were signed in June 2022 and the first delivery was received in May 2025. The contract is expected to last for 15 years. The EU requires some diversity in fuel supply options. In March 2023 CEZ signed an agreement with Westinghouse for the supply of VVER-440 fuel assemblies to Dukovany from 2024.

In April 2025 Kazatomprom signed a deal to supply about one-third of the uranium needed for Westinghouse fuel assemblies used at the Temelin plant.

## Radioactive waste management

The Radioactive Waste Repository Authority (RAWRA or *Správa úložišť radioaktivních odpadů*, SÚRAO) was established in 1997 and is responsible for the disposal of all radioactive waste in the

Czech Republic. There is no state policy on reprocessing and the decision is left to CEZ, which does not perceive it as being economic. However, the question remains open. All producers of radioactive wastes bear the full cost of management and disposal.

**Used fuel, HLW**

CEZ is fully responsible for storage and management of its used fuel until it is handed over to the state organization Radioactive Waste Repository Authority (RAWRA or SURAO). In October 2016 it signed a four-year service contract with Finland's Posiva Solutions to advise in the site selection process for a repository, the disposal concept, and repository design. Finnish engineering company Saanio & Riekkola Oy as well as SKB International and the Geological Survey of Finland are involved.

Used fuel is stored at each power plant. Originally, used fuel from Dukovany was sent to the interim storage facility at the Bohunice plant (now in Slovakia). The dissolution of Czechoslovakia in 1993 meant that this used fuel originating from Dukovany was stored in a different country, and therefore required repatriation[k]. An interim dry storage facility with capacity of 600 t was built at Dukovany, and the plant's used fuel storage pools were reracked to increase capacity. The dry storage facility commenced operation in 1995 and since then another storage facility has been built there.

Reracking of storage pools has also taken place at Temelin, and an interim dry storage facility is in use there. CEZ creates an internal financial reserve for long-term used fuel storage. Skalka in southern Moravia had been considered in the late 1990s for a central used fuel interim storage facility as an alternative to the Temelin storage facility and to the storage capacity expansion at Dukovany (beyond the 600 t facility)[m].

An interim storage facility for used research reactor fuel is located at the Rez nuclear research institute (see section below on Research and development).

Eventual provision of a deep geological repository (DGR) for high-level waste is the responsibility of RAWRA/SURAO. The government originally planned the selection of a candidate and a reserve site by 2025, with construction to start after 2050, and operation to begin in 2065. However, these proposed dates have been brought forward in order to meet the deadline outlined in the EU taxonomy for sustainable activities, which requires deep geological repositories to be operational by 2050 for nuclear energy to be classified as a sustainable investment. The government emphasized that it would continue to put forward the case for the

deadline to be revised. In 2020 the government published a list of four shortlisted sites: Hrádek and Horka, which are in the central region of the country; Březový potok, in the southwest; and Janoch near the Temelín nuclear power plant. The cost of building and operating the deep repository is estimated at CZK 130 billion (€5.2 billion).

**Low and intermediate-level waste**

At the beginning of 2000, ownership of the country's three repositories – Dukovany, Richard and Bratrstvi – was transferred to the state under the management of RAWRA/SURAO. Waste from non-power applications is disposed of at the Richard and Bratrstvi repositories[l].

The Dukovany repository is the largest of the repositories and was built specifically for the disposal of low-level and intermediate-level radioactive waste generated during the operation of the Dukovany and Temelin nuclear plants. The 55,000 $m^3$ storage volume provides enough space for the waste from both plants, even with their operational lifetimes extended to 40 years. Each of the repository's 112 vaults can accommodate about 1600 individual 200-litre drums. It began operation in 1995.

**Funding of wastes and decommissioning**

Under the Atomic Energy Act 2002, CEZ as nuclear plant operator is required to put aside funds for waste disposal, lodging these with the Czech National Bank. The rate is CZK 0.05 (€0.002) per kWh. The Act also requires that nuclear plants are decommissioned following the end of their operating lifetimes and CEZ is also progressively funding this. The adequacy of reserve funds for decommissioning is under the supervision of RAWRA/SURAO.

# Research & development

The Nuclear Research Institute Řež plc (*Ústav jaderného výzkumu Řež a.s.*, ÚJV) was founded in 1955 through an agreement between Czechoslovakia and the Soviet Union, and later incorporated into the Czechoslovak Academy of Sciences. The village of Řež is located in the Husinec municipality in central Bohemia 11 km northwest of Prague. In 1992, the Rez Nuclear Research Institute was privatized[n] and is now owned by CEZ (52.4%) with Slovak Electric (SE, 27.8%), Škoda JS (17.4%) and the Husinec municipality (2.4%)[o].

Two research reactors are currently in operation at Rez: the 10 MWt LVR-15, which went critical in September 1957, and the 5 kWt LR-0 (criticality 1982). The Czech Technical University in Prague operates a third research reactor, the 5 kWt VR-1 Sparrow (criticality 1990). The VR-1 was the first research reactor supplied with highly enriched uranium (HEU) from Russia to convert to low enriched uranium (LEU) fuel[8].

Used HEU fuel that was stored at Rez has been returned to Russia as part of the Russian Research Reactor Fuel Return program[9].

# Regulation, safety and non-proliferation

Licensing, nuclear safety, waste management, safeguards, and radiation protection are regulated by the State Office for Nuclear Safety (SÚJB), which took over these responsibilities from the former Czechoslovak Atomic Energy Commission in 1993. The SUJB reports directly to cabinet.

In November 2013 a peer review team led by the IAEA completed an Integrated Regulatory Review Service (IRRS) mission to the Czech Republic and reported "that the Czech regulatory system for nuclear and radiation safety is robust and that the State Office for Nuclear Safety (SÚJB) is an effective and independent regulatory body." In May 2017 a follow-up IRRS mission reported favourably but said work was still required in implementing the country's new Atomic Act.

The Atomic Energy Act of 1997 was amended in 2002 to harmonize with European Union legislation, and covers all nuclear energy matters. The SUJB was involved in drafting a new Atomic Act 2016 which entered into force in January 2017. It represents a thorough overhaul of Czech nuclear legislation and addresses particularly nuclear security, nuclear non-proliferation and physical protection of nuclear materials.

## Non-proliferation

The Czech Republic is a party to the Nuclear Non-Proliferation Treaty (NPT) since 1993 as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1997. It is member of the Nuclear Suppliers Group and, since May 2004, of Euratom. The Additional Protocol in relation to its safeguards agreements with the International Atomic Energy Agency came into force in 2002.

# Notes & references

## Notes

c. VSB was acquired by Hochtief in 1999, becoming Hochtief VSB, and later Hochtief CZ. [Back]

d. The two VVER-440 reactors at Loviisa in Finland were built in the 1970s incorporating hardwired Western I&C systems, but it wasn't until 2008 that Areva commenced a six-year renewal project to instal modern digital I&C. In addition, Western systems were incorporated into the Loviisa plant at the design stage, whereas it was not until the Temelin reactors were substantially built that Western I&C was introduced. [Back]

e. In 1992, the state's National Property Fund founded ČEZ, a. s. from the state enterprise *České energetické závody* (Czech power plants). In 1993, 27% of CEZ shares were privatized in the first wave of 'voucher privatization'. The second wave was in 1995, when a further 6% was sold. In 2001, the government commenced the process of choosing a strategic investor for the National Property Fund's remaining stake in CEZ and six regional distribution companies, but this was cancelled at the beginning of 2002. In the mid-2000s, the government said it intended to sell a 16% stake in CEZ, leaving it with 51%, and in 2007 commenced the sale of a 7% stake, but this was not completed and at the same time CEZ was repurchasing shares on the open market (thus increasing the government stake). By the end of 2009, the government owned just under 70% of CEZ, with asset managers holding about 21% and about 5% held by private individuals. (The banks UniCredit Bank Czech, Citibank Europe and ČSOB own almost all of the shares held by asset managers.) [Back]

f. In October 2009, the consortium of Skoda JS, Atomstroyexport and Gidropress submitted qualification documents for VVER-1200 units for Temelin 3&4. The model was represented as the Europe-tailored reactor design, the MIR-1200 (Modernized International Reactor), also intended for the Baltic plant in Kaliningrad and bid for Turkey and Finland, and similar to the plants under construction at Leningrad II and Novovoronezh II. CEZ became the 100% owner of Skoda JS in November 2022. [Back]

g. In 2009, the Jess joint venture was established between Slovak state-owned decommissioning company Javys (51%) and CEZ (49%) to construct a new nuclear unit at the Bohunice site in Slovakia. See Slovakian nuclear JV gets government blessing, World Nuclear News (10 December 2009) [Back]

h. In a May 2007 decree, the Czech Republic government approved the continuation of uranium mining and processing at the Rožná deposit for as long as it remains economic. The government required that a new proposal on mining activity at Rožná, based on results of geological exploration of reserves, economic and energy security aspects, be submitted by June 2012.[4] [Back]

i. In situ leach (ISL) mining took place at Diamo's northern mine of Stráž pod Ralskem from 1967 until 1996. Since 1996, uranium has been exploited as a by-product of the remediation of the Stráž uranium deposit. [Back]

k. Used fuel from Dukovany was sent to the Bohunice interim storage facility prior to being sent to Russia. However, in 1993, Russia decided to only accept used fuel for reprocessing, not direct disposal, and Slovakia said it would no longer accept used fuel from Dukovany. [Back]

l. Disposal of low-level institutional waste has been carried since 1964 out at the former Richard limestone mine near the town of Litoměřice, Northern Bohemia. (So-called institutional waste is generated in the health sector, industry, research and agriculture. Examples of such waste include disused meters and radionuclide emitters, contaminated clothing, cloth, paper and syringes.) The total volume of the Richard facility exceeds 17,000 $m^3$, about half of which comprises disposal capacity. Assuming the repository continues to be filled at the current pace of 100 to 200 drums per year, its operational life will continue up to 2070. Institutional waste containing naturally occurring radioactive material (NORM) is disposed of at the Bratrství repository, near the town of Jáchymov. The facility was constructed from one of the mined cavities of a former uranium mine and has a disposal capacity of approximately 1200 $m^3$. It was put into operation in 1974 and was 80% full at the end of 2008.

Another repository at Hostím near the city of Beroun was closed down in 1965. It was constructed in a former limestone mine and put into operation in 1959 for institutional radioactive waste. Following its closure, most of waste packages were transferred to the Richard repository. The repository was filled with concrete and sealed in 1997. The site is monitored by RAWRA. [Back]

m. At the end of 1999, the Ministry of Environment rejected the plan to construct a central interim used fuel storage facility (of 2900 t capacity) at Skalka on the grounds that approval had been given to the expansion of storage capacity at Dukovany. A year later, CEZ submitted a proposal to construct a repository at Skalka, and the site was reserved. However, site selection for a deep geological waste repository from 2014 appears to exclude it. [Back]

n. The Řež Nuclear Research Institute (ÚJV) underwent restructuring in 1972, when the bulk of the organization was incorporated into the Czechoslovak Atomic Energy Commission, which was at the time responsible for the nuclear programme development. At the end of December 1992 the Institute was privatized as a joint-stock company called Nuclear Research Institute Rez. The Institute began offering designing and engineering services following the November 2002 acquisition of the design division of Energoprojekt Praha. In 2002 an R&D focused subsidiary, Research Centre Rez, was formed. In 2011 the UJV Group was established, and the name of the company was changed to UJV Rez in 2012. The UJV Group consists of the parent company and three 100% owned subsidiaries: Research Centre Rez; Skoda Praha; and Research and Testing Institute Pilsen. [Back]

o. Slovak Electric, (*Slovenské Elektrárne*, SE) is 66%-owned by Italy's Enel (the rest of SE is held by the Slovak government); as of December 2022, Skoda JS is 100% owned by CEZ. [Back]

## References


1. Temelin's hard-earned success, Nuclear Engineering International (3 July 2002) [Back]
2. Nuclear power plants in 2009, ČEZ a. s. Generation Division (2010)
3. CEZ opened the public tender for a contactor of nuclear units, CEZ Group press release (3 August 2009) [Back]
4. Prolongation of uranium mining on Rožná deposit In Dolní Rožínka location, Ministry of Industry and Trade of the Czech Republic press release (28 May 2007) [Back]
5. Discussions Continue Over Rozna, Uran Limited announcement (27 February 2007) [Back]
6. Political Support for Increased Nuclear Power and Uranium Mining in Czech Republic. Appeals Lodged Over Exploration Permits in Czech Republic, Uran Limited announcement (16 June 2008) [Back]

7. Temelin to use only Russian fuel from 2010, World Nuclear News (20 October 2009); TVEL provides Skoda with test fuel assembly, World Nuclear News (22 November 2007) [Back]
8. NNSA Completes Czech Research Reactor Conversion, National Nuclear Security Administration, US Department of Energy, press release (4 November 2005) [Back]
9. Rez research reactor fuel returned to Russia, World Nuclear News (10 December 2007) [Back]
10. New Raw Material Policy for Minerals and Their Resources, Ministry of Industry and Trade (7 November 2017) [Back]

### General sources

CEZ website (www.cez.cz)
Diamo website (www.diamo.cz)
Radioactive Waste Repository Authority (SÚRAO) website (www.surao.cz)
State Office for Nuclear Safety (SÚJB) website (www.sujb.cz)
Country Nuclear Power Profiles: Czech Republic, International Atomic Energy Agency
The Source Book on Soviet-Designed Nuclear Power Plants, Nuclear Energy Institute
*Energy in East Europe*, 23 January 2004

## Related information

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